Proof Problems for the Alignment Kernel

TB Hardware Accelerator project

Our project lines up two strings of DNA letters to find where they differ. The standard method fills in a whole table. Our design first guesses where the answer lies (the *vote*) and then fills in only a narrow strip of the table (the *window*). The statements below say how much work this saves and when it is guaranteed to give the same answer.

We have draft proofs, and nobody has checked them yet. **Please prove these by hand, without looking at the drafts.** Afterwards we will compare. If you find a statement that is false or needs an extra condition, that is the most useful result of all.

Take a whole part, not single problems. The problems inside a part build on each other: the early ones are the steps and the last one is the goal. Section 2 shows how the parts connect.

Difficulty: $\star$ counting or a few lines, $\star\star$ a real argument, $\star\star\star$ longer, usually induction. Write up each part as one piece, with the part letter and your names on it.

1 The rules

- The *read* is $a = a_1 \dots a_m$ and the *reference* is $b = b_1 \dots b_n$. Letters are A, C, G, T.
- Scores: a match scores $M > 0$, a mismatch scores $x \le M$, and each gapped letter scores $g < 0$. The project uses $M = 2$, $x = -1$, $g = -2$, $m = 150$, $n \le 180$.
- $s(a_i, b_j) = M$ if $a_i = b_j$, and x otherwise.
- *Nodes* are pairs (i, j) with $0 \le i \le m$ and $0 \le j \le n$. A node with $i, j \ge 1$ is a *cell*. The *diagonal* of a node is $j - i$.
- A *path* is a sequence of steps. A diagonal step $(i-1, j-1) \to (i, j)$ scores $s(a_i, b_j)$. A down step $(i-1, j) \to (i, j)$ and a right step $(i, j-1) \to (i, j)$ each score g; these are *gap steps*. The score of a path is the sum of its steps. The empty path scores 0.

The table. $H(i, 0) = H(0, j) = 0$, and for $i, j \ge 1$

$$H(i,j) = \max\big\{\, 0,\ H(i-1,j-1) + s(a_i, b_j),\ H(i-1,j) + g,\ H(i,j-1) + g \,\big\}. \tag{1}$$

The answer is $S = \max_{i,j} H(i, j)$.

A small example. Reference GATTACA, read GATACA, with $M = 2$, $x = -1$, $g = -2$:

	G	A	T	T	A	C	A
G	2	0	0	0	0	0	0
A	0	4	2	0	2	0	2
T	0	2	6	4	2	1	0
A	0	2	4	5	6	4	3
C	0	0	2	3	4	8	6
A	0	2	0	1	5	6	10

Here $S = 10$. The best path uses one gap step: the read is missing one T.

2 How the parts connect

Part	What it establishes	Problems	Needs	Level
A	What the numbers in the table mean, and how big they can get	1–4	only the rules	$\star\star\star$
B	The score rewritten as a cost	5	only the rules	$\star\star$
C	When the window gives the same answer as the full table	6–10	Problem 1 from Part A	$\star\star$
D	When the vote is guaranteed to pick a good place	11–15	Problems 7 and 8 from Part C	$\star\star\star$
E	How much hardware and time each design needs	16	nothing	$\star$
F	Two separate bounds	17–18	nothing	$\star\star$

Parts A, B, E and F can each be done alone. Part C should be done by people who have read Problem 1. Part D should be done after Part C, or by a group that is willing to take Problems 7 and 8 as given. If you use a result from another part, quote it by its problem number; you do not have to prove it again.

3 Part A: the full table

Goal. Problems 2 and 4: how large a score can be, and how few bits a cell needs. **Chain.** $1 \to 2$ and $3 \to 4$. Problem 1 says what H means; Problem 3 is the long induction.

Problem 1 ($\star\star$)**.** Prove that $H(i, j)$ is the maximum score over all paths that end at the node (i, j), the empty path included.

Problem 2 ($\star$)**.** Prove that $0 \le H(i, j) \le M \min(i, j)$. Deduce that $S \le M \min(m, n)$ and that S fits in $\lceil \log_2(M \min(m, n) + 1) \rceil$ bits. Evaluate this for the project values.

Problem 3 ($\star\star\star$)**.** Prove that for all $1 \le i \le m$ and $1 \le j \le n$:

$$
\begin{aligned}
g &\le H(i, j) - H(i, j-1) \le M - g, \\
g &\le H(i, j) - H(i-1, j) \le M - g, \\
x &\le H(i, j) - H(i-1, j-1) \le M.
\end{aligned}
$$

Problem 4 ($\star$)**.** Using Problem 3, how many different values can a side-by-side difference take? A same-diagonal difference? How many bits does each need? Evaluate for the project values.

4 Part B: score as a cost

For a path P from (i_0, j_0) to (i_1, j_1) let μ, ν, γ be its numbers of matches, mismatches and gap steps, and let $u = i_0 + (m - i_1) + j_0 + (n - j_1)$ be the number of letters it leaves unaligned. Define

$$
c(P) = (M - x)\,\nu + \Big(\frac{M}{2} - g\Big)\gamma + \frac{M}{2}\,u.
$$

Goal. Problem 5, a single identity. It uses nothing except the rules.

Problem 5 ($\star\star$)**.** Prove that $\text{score}(P) = \frac{M}{2}(m+n) - c(P)$ for every path P. Deduce that $S = \frac{M}{2}(m+n) - \min_P c(P)$. What are the three cost coefficients for the project values?

5 Part C: the window

Fix an integer d_0 and a half-width $w \geq 0$, and let $W = 2w + 1$. The *window* is the set of cells

$$\mathcal{W} = \big\{(i,j) : 1 \leq i \leq m,\ 1 \leq j \leq n,\ |j - i - d_0| \leq w\big\}.$$

$H_{\mathcal{W}}$ is defined by the recurrence (1) on the cells of $\mathcal{W}$ and is 0 at every other node. Let $S_{\mathcal{W}} = \max H_{\mathcal{W}}$. A path is *admissible* if every node on it, except possibly its start node, is a cell of $\mathcal{W}$. **Goal.** Problem 10: if the best path fits in the window, the window gives the same answer. **Chain.** Problem 6 stands alone and says how much work is saved. Then $7 \to 8$, and $7, 8, 9 \to 10$. Problem 7 is the window version of Problem 1 and can follow the same pattern.

Problem 6 ($\star$)**.** Prove that $|\mathcal{W}| \leq mW$. What fraction of the full table is this for $m = 150$, $n = 180$, $w = 8$?

Problem 7 ($\star\star$)**.** Prove that for a cell $(i,j) \in \mathcal{W}$, $H_{\mathcal{W}}(i,j)$ is the maximum score over admissible paths ending at (i,j), the empty path included.

Problem 8 ($\star$)**.** Prove that $S_{\mathcal{W}} \leq S$.

Problem 9 ($\star\star$)**.** Prove that the diagonals of the nodes of a path form a set of consecutive integers $\{k_{\text{lo}}, \ldots, k_{\text{hi}}\}$, and that $k_{\text{hi}} - k_{\text{lo}}$ is at most the number of gap steps of the path.

Problem 10 ($\star\star$)**.** Suppose $S > 0$ and some path P^* with $\text{score}(P^*) = S$ has diagonal range $[k_{\text{lo}}, k_{\text{hi}}]$ with $k_{\text{hi}} - w \leq d_0 \leq k_{\text{lo}} + w$. Prove that $S_{\mathcal{W}} = S$. Show that the condition holds whenever d_0 is one of the diagonals of P^* and P^* has at most w gap steps.

6 Part D: the vote

In this part indices start at 0: the read is $r_0 \ldots r_{m-1}$, the window of the reference is $b_0 \ldots b_{n-1}$, and diagonal k pairs r_i with b_{i+k}. Fix $\lambda \geq 0$. For $-(m-1) \leq k \leq n-1$ let

$$I(k) = \{\, i : 0 \leq i < m,\ 0 \leq i + k < n \,\}, \qquad N(k) = |I(k)|,$$
$$V_r(k) = \big|\{\, i \in I(k) : r_i = b_{i+k} \,\}\big|, \qquad Z_r(k) = (1+\lambda)V_r(k) - \lambda N(k).$$

The vote returns a diagonal d_0 that maximises Z_r.

Read model. Let E be the genome around the window, with $b_j = E[\omega + j]$. A *template* $t_0 \ldots t_{m-1}$ has a start p and at most one indel of length g at read position q:

- no indel: $t_i = E[p + i]$;
- deletion: $t_i = E[p + i]$ for $i < q$, and $t_i = E[p + g + i]$ for $i \geq q$;
- insertion: $t_i = E[p + i]$ for $i < q$; t_i is unconstrained for $q \leq i < q + g$; and $t_i = E[p - g + i]$ for $i \geq q + g$.

F is the set of unconstrained positions. A read r *follows the template with at most e substitutions* if $r_i = t_i$ for all $i \notin F$ except at most e positions.

The *true diagonals* are $k_1 = p - \omega$ and $k_2 = k_1$ (no indel), $k_1 + g$ (deletion) or $k_1 - g$ (insertion). Put $\kappa_{\text{lo}} = \min(k_1, k_2)$, $\kappa_{\text{hi}} = \max(k_1, k_2)$. The *true alignment* T pairs each read letter with the

reference letter it was copied from. The set of *acceptable* diagonals is $\mathcal{A} = \{d : \kappa_{\mathrm{hi}} - w \le d \le \kappa_{\mathrm{lo}} + w\}$. Define

$$V^-(k) = \big|\{\, i \in I(k) \setminus F : t_i = b_{i+k} \,\}\big|, \qquad V^+(k) = V^-(k) + |I(k) \cap F|,$$

$Z^{\pm}(k) = (1+\lambda)V^{\pm}(k) - \lambda N(k)$, and the *margin*

$$\Delta_\lambda = \max_{k \in \mathcal{A}} Z^-(k) - \max_{k \notin \mathcal{A}} Z^+(k).$$

Goal. Problem 13: if the read has few enough substitutions, the window chosen by the vote contains the true alignment. **Chain.** 11 → 12 → 13, and Problem 13 also uses Problems 7 and 8. Problem 14 follows from 12. Problem 15 is a separate count. Read the definitions above twice before starting; they are the hard part.

Problem 11 (⋆⋆)**.** Prove that if r follows the template with at most e substitutions, then $V^-(k) - e \le V_r(k) \le V^+(k) + e$ for every k.

Problem 12 (⋆⋆⋆)**.** Prove: if r follows the template with at most e substitutions and $2(1+\lambda)e < \Delta_\lambda$, then every maximiser of Z_r is acceptable.

Problem 13 (⋆⋆)**.** Under the hypothesis of Problem 12, prove that the window centred on d_0 contains every diagonal of the true alignment T, and that $\mathrm{score}(T) \le S_{\mathcal{W}} \le S$.

Problem 14 (⋆⋆)**.** Suppose each read letter is substituted independently with probability ε, and the template tolerates e^* substitutions in the sense of Problem 12. Prove that the probability that the vote is not acceptable is at most $\Pr[\mathrm{Bin}(m, \varepsilon) > e^*]$. Evaluate it for $m = 150$, $e^* = 19$, $\varepsilon = 0.02$.

Problem 15 (⋆)**.** Prove that computing $V_r(k)$ for every k by direct comparison uses exactly mn comparisons of letters.

7 Part E: counting the hardware

A machine that fills the full table one anti-diagonal per clock tick uses $\min(m, n)$ cells and $m+n-1$ ticks, and stores $2mn$ bits of direction pointers. The window machine uses W cells and m ticks and stores $2mW$ bits. The vote takes $m + n - 1$ ticks.
Goal. Problem 16, which is counting only. Part (c) is the point of it.

Problem 16 (⋆)**.** (a) Evaluate all of these for $m = 150$, $n = 180$, $W = 17$. (b) A read goes through the vote and then the window. Is that more or fewer ticks than the full-table machine? Prove your answer for all $m, n \ge 1$. (c) State precisely what the new design saves and what it does not.

8 Part F: two side results

Problems 17 and 18 do not depend on each other or on anything else.

Problem 17 (⋆⋆)**.** Split the read into consecutive segments of length at most S_{seg}. Segment t votes for a diagonal d_t. A row in segment t computes the diagonals from $\min(d_{t-1}, d_t, d_{t+1}) - w$ to $\max(d_{t-1}, d_t, d_{t+1}) + w$ (a missing neighbour is left out). Let $G = \sum_t |d_{t+1} - d_t|$. Prove that at most $mW + 2S_{\mathrm{seg}}G$ cells are computed.

Problem 18 (⋆)**.** A read skips the window with probability ρ. The vote costs C_{vote} and the window costs C_{SW}. Prove that the expected cost per read is $C_{\mathrm{vote}} + (1 - \rho)\, C_{\mathrm{SW}}$.

9 Part G: open questions

We do not have proofs of these. We do not know whether they are true.

Q1. Under the read model of Part D, is $S_{\mathcal{W}} = S$? Problem 13 gives only $\text{score}(T) \le S_{\mathcal{W}} \le S$. Find a proof, an extra condition that makes it true, or a counterexample.

Q2. The condition $2(1 + \lambda)e < \Delta_\lambda$ in Problem 12 assumes the worst case for every substitution. Can it be weakened?

Q3. With $\lambda = 0$ the margin can be negative for our sequences. Is there a simple property of the reference that guarantees a positive margin?

Q4. Extend Problem 12 to reads with two indels.

Sign-up

Part	Problems	Names	Written	Checked by
A	1–4			
B	5			
C	6–10			
D	11–15			
E	16			
F	17–18			
G	open questions			